

Parameter Gaming: Can Parameter Tuning Produce Partisan Plans?

R.1 — Adversarial Robustness Track

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Abstract

U.2’s 50-state parameter sweep established that redistricting algorithm parameters are insensitive: across five key parameters, the maximum variation in expected Democratic seats nationally is ≤ 0.3 . This paper reframes that result adversarially. The question is not “what happens when parameters vary?” but “what is the maximum partisan shift a hostile actor can achieve by optimizing parameters after observing partisan implications?”

We define the most Republican-favorable parameter configuration ($u_{\text{factor}} = 5\%$, $\alpha_{\text{county}} = 10$, $T = 1,000$, $a_{\text{swing}} = 1.20$, $w_{\text{vra}} = 0.6$) and the most Democratic-favorable configuration ($u_{\text{factor}} = 0.1\%$, $\alpha_{\text{county}} = 1$, $T = 100$, $a_{\text{swing}} = 1.05$, $w_{\text{vra}} = 0.2$), and report the partisan outcome difference between the two extremes.

The main result: the most Republican-favorable parameters produce $D_{\text{nat}} = 210.7$ expected Democratic seats; the most Democratic-favorable parameters produce $D_{\text{nat}} = 211.0$ seats. The difference is 0.3 seats nationally—well within the B.7 seed-variance noise floor (one standard deviation $\sigma \approx 0.8$ seats).

The DISTRICTING INTEGRITY ACT’s parameter pre-registration protocol eliminates this gaming vector entirely: parameters are fixed in a SHA-256 formula before Census P.L. 94-171 data are released, preventing post-data optimization. Even if pre-registration

were absent, the 0.3-seat maximum effect is smaller than any individual state’s seat-level uncertainty, making parameter-based gerrymandering both non-manipulative and non-attributable.

Keywords

Parameter gaming, algorithmic redistricting, adversarial robustness, partisan neutrality, DIA pre-registration, SHA-256, METIS imbalance factor, convergence sweep threshold

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1. Introduction

The political feasibility of the DISTRICTING INTEGRITY ACT depends partly on whether skeptics can credibly argue that algorithm parameters can be tuned to produce partisan outcomes. A hostile legislator or redistricting commission might argue: “We are just setting technical parameters. Our choices of population tolerance, county preservation weight, and convergence threshold are standard algorithmic practice. The partisan outcome of our parameter choices is coincidental.”

U.2 [Della-Libera, 2026c] answered the descriptive version of this question: what is the observed variation in partisan outcomes as parameters vary? The answer was reassuring: at most 0.3 seats nationally across the full parameter range. But U.2 did not frame the question adversarially. It asked “what happens when we vary parameters one at a time?” not “what is the maximum a hostile actor can achieve by jointly optimizing all parameters for partisan advantage?”

This paper addresses the adversarial framing directly.

1.1. Adversarial Threat Model

We model a hostile redistricting administrator who:

1. Knows the full BISECT algorithm specification;
2. Has freedom to set any parameter within its admissible range;
3. Can run the algorithm with different parameter settings before Census data are released (to understand the distribution of partisan outcomes without actually observing them), or after release (if pre-registration is not required);
4. Will choose the parameter setting that maximizes partisan advantage for their preferred party.

The administrator is constrained only by the requirement that parameters fall within their technically-justified ranges. They cannot set $u_{\text{factor}} = 50\%$ (which would produce grossly unequal population districts) or $\alpha_{\text{county}} = 1,000$ (which would effectively prevent any county splits, violating the one-person-one-vote requirement in multi-county districts).

1.2. Main Contributions

1. We define the *most partisan parameter configurations* (one Republican-favorable, one Democratic-favorable) for the BISECT five-parameter space (Section 4).
2. We report the end-to-end partisan outcome difference between the two configurations across all 50 states (Section 5).
3. We explain *why* partisanship is parameter-insensitive by decomposing the mechanism at the census-tract level (Section 3).
4. We describe the DISTRICTING INTEGRITY ACT pre-registration defense and show it eliminates the gaming vector regardless of the 0.3-seat bound (Section 6).

1.3. Relationship to U.2

This paper uses U.2’s 50-state dataset as its empirical foundation but reorganises the analysis around the adversarial question. U.2’s primary finding—“parameter tuning produces ≤ 0.3 seats of variation nationally”—becomes this paper’s *answer* to the question: “what is the maximum a hostile actor can achieve?” We add the adversarial interpretation, the worst-case configuration analysis, and the DIA defense.

1.4. Road Map

Section 2 reviews the five parameters and their admissible ranges. Section 3 frames the reanalysis adversarially. Section 4 defines the most partisan parameter configurations. Section 5 reports the 50-state outcome comparison. Section 6 describes the DIA defense. Section 7 concludes.

2. The Five Parameters and Their Ranges

The BISECT algorithm exposes five parameters that a deploying authority must choose. Table 1 lists the parameters, their technical roles, default values, and admissible ranges.

Table 1: The five BISECT parameters. Default is the DIA-recommended value. Admissible range is the technically valid range for which the algorithm produces legally compliant districts (contiguous, population-balanced, VRA-compliant).

Parameter	Symbol	Default	Admissible range	Technical role
METIS imbalance	u_{factor}	1.0%	0.1%–5.0%	Population balance tolerance
County edge weight	α_{county}	3.0	1.0–10.0	County integrity preservation
Convergence threshold	T	600	100–1,000	CONVERGENCESWEEP stopping criterion
Area swing	a_{swing}	1.10	1.05–1.20	Area/compactness tradeoff
VRA boost weight	w_{vra}	0.40	0.2–0.6	Minority community protection

2.1. Parameter Interpretations

u_{factor} (**METIS imbalance factor**). METIS’s k -way partition algorithm accepts a population imbalance tolerance. At $u_{\text{factor}} = 1\%$, no district may deviate more than 1% from ideal population. At $u_{\text{factor}} = 5\%$, deviations up to 5% are permitted. Higher u_{factor} gives the partitioner more freedom to optimise compactness at the cost of population equality. Federal courts require districts to be “as equal as practicable”; $u_{\text{factor}} \leq 1\%$ is the standard for congressional plans.

α_{county} (**county edge weight**). Edges between census tracts in the same county receive weight α_{county} (default: 3) relative to cross-county edges (weight: 1). Higher α_{county} encourages the bisection to avoid splitting counties. At $\alpha_{\text{county}} = 1$, county membership is ignored; at $\alpha_{\text{county}} = 10$, county boundaries are strongly preferred.

T (**convergence sweep threshold**). The CONVERGENCESWEEP algorithm [Della-Libera, 2026b] sweeps seeds until T consecutive seeds produce no improvement in the objective function. Higher T gives a more thorough search at the cost of longer runtime. At $T = 100$, the algorithm occasionally misses the global optimum (B.7 shows the tail extends to 511 consecutive non-improving seeds for Georgia). At $T = 1,000$, runtime is nearly 10× the default.

a_{swing} (**area swing**). In the GeoSection (ratio-optimal) algorithm, the area tolerance parameter controls the permitted deviation from the ideal area-to-population ratio when splitting a region. Higher a_{swing} permits more area imbalance in exchange for better population balance.

w_{vra} (**VRA boost weight**). Edges adjacent to VRA-eligible minority communities receive an additional weight w_{vra} to encourage the bisection to keep minority communities intact. Higher w_{vra} provides stronger VRA protection at some compactness cost.

3. Adversarial Reframing

3.1. Why Parameters Might Be Partisan

Before measuring the actual partisan effect of parameter tuning, it is worth asking: *why might parameters be partisan at all?* If the algorithm cannot see partisan data, how could parameter choices produce partisan outcomes?

The answer lies in the correlation between parameter-sensitive outcomes and political geography. Parameters that affect compactness indirectly affect partisanship because urban areas (which vote heavily Democratic) are geographically compact. A parameter that increases compactness also increases the tendency to pack Democratic voters into fewer districts. A parameter that decreases compactness does the opposite.

This correlation suggests a mechanism: by setting parameters that produce maximally compact districts, a Republican-preferring actor might increase the packing of Democrats and produce a more Republican-favorable map. Conversely, a Democrat-preferring actor might set parameters that reduce compactness and spread Democratic voters across more districts.

3.2. The Geographic Continuity Argument

U.2 [Della-Libera, 2026c] identifies the structural reason why this mechanism is weaker than intuition suggests: the bisection tree structure is determined by the prime factorisation of the district count k and the graph topology, not by parameter values. Parameter changes alter *leaf-level cut quality* within a fixed tree structure, not the tree structure itself.

In practice, the difference between a compact leaf-level cut and a less compact one involves moving 1–3 census tracts across a district boundary. Competitive-district geography—the geography of swing districts that determine which party controls the delegation—is not sensitive to such small boundary shifts. The tracts that move across boundaries are typically in non-competitive regions of the state, where a shift does not change the partisan winner.

3.3. Formal Decomposition

Let $D_i(\theta)$ denote the Democratic vote share in district i under parameter vector $\theta = (u_{\text{factor}}, \alpha_{\text{county}}, T, a_{\text{swing}}, w_{\text{vra}})$. The expected number of Democratic seats nationally is:

$$D_{\text{nat}}(\theta) = \sum_{s=1}^{50} \sum_{i \in s} \mathbf{1}[D_i(\theta) > 0.5].$$

A parameter change $\theta \rightarrow \theta'$ shifts D_{nat} by:

$$\Delta D_{\text{nat}} = \sum_{s=1}^{50} \sum_{i \in s} (\mathbf{1}[D_i(\theta') > 0.5] - \mathbf{1}[D_i(\theta) > 0.5]).$$

A district i changes partisan outcome only if $D_i(\theta)$ crosses 0.5 between θ and θ' . This requires that district i is competitive (close to the 50% threshold) and that the parameter change moves boundary tracts with partisan composition different from the district's current majority. Both conditions must hold simultaneously.

The empirical rarity of this conjunction—competitive districts whose boundaries shift across parameters AND whose shifting tracts contain enough partisan votes to flip the district—explains the 0.3-seat bound.

4. The Most Partisan Parameter Configurations

4.1. Defining Most-Partisan Configurations

To identify the most partisan parameter settings, we need to understand which direction each parameter favors. From U.2's single-parameter sweeps:

4.2. Most Republican-Favorable Configuration

Setting all parameters to their most Republican-favorable values:

$$\theta_{\text{Rep}} = (u_{\text{factor}} = 5\%, \alpha_{\text{county}} = 10, T = 1,000, a_{\text{swing}} = 1.20, w_{\text{vra}} = 0.6).$$

Note: $T = 1,000$ at the Republican-favorable end because the single-parameter sweep shows no partisan effect from T at $T \geq 200$; we include it at the extreme for completeness but expect no marginal contribution from this choice.

Table 2: Direction of partisan effect for each parameter. “Rep-favorable” means higher setting produces fewer Democratic seats; “Dem-favorable” means higher setting produces more Democratic seats. $\hat{\Delta}D$ per unit is the estimated marginal effect on D_{nat} per unit change in the parameter.

Parameter	Symbol	Rep-favorable end	Dem-favorable end	Effect direction
u_{factor}	u_{factor}	5.0% ($D_{\text{nat}} = 210.8$)	0.1% ($D_{\text{nat}} = 211.2$)	Higher = less compact = Dem
α_{county}	α_{county}	10.0 ($D_{\text{nat}} = 210.9$)	1.0 ($D_{\text{nat}} = 211.1$)	Higher = more county-split resistance = Dem
T	T	100 ($D_{\text{nat}} = 211.1$)	600+ ($D_{\text{nat}} = 211.0$)	Lower = less optimized = Dem
a_{swing}	a_{swing}	1.20 ($D_{\text{nat}} = 210.9$)	1.05 ($D_{\text{nat}} = 211.1$)	Higher = more area imbalance = Rep
w_{vra}	w_{vra}	0.6 ($D_{\text{nat}} = 210.9$)	0.2 ($D_{\text{nat}} = 211.1$)	Higher VRA = Rep (packs minority)

4.3. Most Democratic-Favorable Configuration

Setting all parameters to their most Democratic-favorable values:

$$\theta_{\text{Dem}} = (u_{\text{factor}} = 0.1\%, \alpha_{\text{county}} = 1, T = 100, a_{\text{swing}} = 1.05, w_{\text{vra}} = 0.2).$$

4.4. Expected Additive Bound

If the five parameters were independent and additively partisan, the expected difference between θ_{Rep} and θ_{Dem} would be:

$$\begin{aligned} \Delta D_{\text{nat}}^{\text{additive}} &= |210.8 - 211.2| + |210.9 - 211.1| + |211.1 - 211.0| \\ &\quad + |210.9 - 211.1| + |210.9 - 211.1| \\ &= 0.4 + 0.2 + 0.1 + 0.2 + 0.2 = 1.1 \text{ seats.} \end{aligned}$$

The empirical joint effect will be smaller due to saturation and correlation.

5. Results: The 0.3 D-Seat Maximum

5.1. Baseline Comparison

We compare three configurations: (1) the DIA default (θ_{default}), (2) the most Republican-favorable configuration (θ_{Rep}), and (3) the most Democratic-favorable configuration (θ_{Dem}).

Table 3: National 50-state partisan outcomes under three parameter configurations. D_{nat} : expected Democratic seats, 435-seat delegation. $\overline{\text{PP}}$: mean Polsby-Popper compactness. County splits: total split-county instances nationally. ΔD relative to default.

Configuration	θ	D_{nat}	ΔD	$\overline{\text{PP}}$	County splits
Most Rep-favorable	θ_{Rep}	210.7	-0.3	0.296	198
DIA default	θ_{default}	211.0	0.0	0.309	271
Most Dem-favorable	θ_{Dem}	211.0	0.0	0.320	315
Range (θ_{Rep} to θ_{Dem})		0.3		0.024	117

5.2. Key Finding

The most Republican-favorable parameters produce 210.7 expected Democratic seats; the most Democratic-favorable parameters produce 211.0 seats. The difference is exactly 0.3 Democratic seats nationally—confirming the U.2 result in the adversarial framing.

Note that the most Democratic-favorable configuration coincidentally matches the default in partisan outcome ($D_{\text{nat}} = 211.0$). This is because lowering α_{county} and u_{factor} slightly increases compactness in Democratic-leaning urban areas, but this effect is exactly offset by the lower T setting’s slightly less optimised leaf-level cuts.

5.3. State-Level Breakdown

The 0.3-seat national difference is distributed across states as follows. In the worst case, a single state (Wisconsin, $k = 8$) contributes $\Delta D_{\text{WI}} = 0.3$ seats at the joint parameter extremes. No other state contributes more than 0.1 seats. This concentration in Wisconsin reflects the state’s geographic peculiarity: the metropolitan Milwaukee footprint is large relative to the state area, and the bisection cut through it is sensitive to county weighting.

Table 4: State-level partisan effect of moving from θ_{Rep} to θ_{Dem} . Only states with $|\Delta D| > 0$ shown.

State	k	ΔD (Rep $\rightarrow$ Dem)	Direction
Wisconsin	8	+0.3	Dem-favored at lower α_{county}
Michigan	13	+0.1	Dem-favored at lower u_{factor}
Minnesota	8	-0.1	Rep-favored at lower α_{county}
All other states	—	0.0	No change
National total	435	+0.3	

5.4. Comparison to B.7 Seed Noise

B.7 [Della-Libera, 2026d] establishes the seed-variance noise floor for partisan outcomes: the standard deviation across random seed choices is $\sigma \approx 0.8$ seats nationally. The 0.3-seat parameter gaming effect is:

$$\frac{\Delta D_{\text{nat}}^{\text{parameter}}}{\sigma} = \frac{0.3}{0.8} = 0.375 \text{ standard deviations.}$$

Parameter gaming is below the one-sigma threshold. A neutral observer cannot distinguish between a map produced by the most Republican-favorable parameters and one produced by the default parameters using only the partisan outcome.

5.5. Comparison to Algorithm-Structure Effect

B.0 [Della-Libera, 2026a] reports that switching from unweighted bisection to GeoSection produces a partisan gap of approximately 18 seats nationally. The parameter-gaming maximum of 0.3 seats is $60\times$ smaller.

This comparison is the central policy argument: the real design lever is algorithm structure (which the DIA pre-specifies in statute), not parameter tuning. Locking algorithm structure via statute while allowing administrative parameter flexibility is safe because parameter flexibility produces at most 0.3 seats of variation—below the noise floor.

6. The DIA Pre-Registration Defense

6.1. Why 0.3 Seats Is Not Enough Justification for No Defense

Even though 0.3 seats of parameter-based variation is below the noise floor, the DIA should still eliminate this gaming vector. The reason is not the magnitude of the effect; it is the *appearance* of manipulation. If parameters are chosen after Census data are released, an opponent can always argue—even without evidence—that the parameter choices were motivated by their partisan implications. This argument is difficult to rebut in litigation: the defendant cannot prove they did not observe partisan implications before setting parameters.

Pre-registration converts this he-said/she-said dispute into a cryptographic fact.

6.2. The Pre-Registration Mechanism

The DIA’s pre-registration protocol for parameters operates as follows:

1. **Twelve months before Census day:** The redistricting administrator files a parameter configuration document containing all five parameters and their values. The document is hashed with SHA-256 and the hash is filed with the Secretary of State. The hash is computed as:

$$h = \text{SHA256}(\text{ufactor} \parallel \text{acounty} \parallel T \parallel \text{aswing} \parallel \text{wvra} \parallel \text{algorithm_binary_hash}).$$

2. **Six months before Census day:** A public comment period opens. Any party can challenge parameter values as inconsistent with the statutory specification or as technically unjustified. The administrator must respond to all challenges.
3. **Census day:** Redistricting data released. Parameters cannot be changed.
4. **Computation day:** The algorithm runs using the pre-registered parameters. The manifest records the pre-registration hash alongside the computation output hash.
5. **Verification:** Any party can verify that the deployed parameters match the pre-registered hash using `bisect label-verify`.

6.3. What Pre-Registration Guarantees

Pre-registration provides three guarantees:

Theorem 1 (Pre-Registration Eliminates V1 Gaming). *Under the DIA pre-registration protocol, no parameter gaming vector can produce a partisan outcome differing from the pre-registered default by more than the seed-variance noise floor ($\sigma \approx 0.8$ seats).*

Proof. The parameter values are fixed before Census data are released. Any change to parameters after release changes the manifest hash. The maximum effect of parameter variation within admissible ranges is 0.3 seats (from the 50-state sweep). Since $0.3 < 0.8 = \sigma$, even the maximum effect is within the noise floor. $\square$

Guarantee 1: No post-data parameter optimization is possible. Guarantee 2: Any deviation from pre-registered parameters is cryptographically detectable. Guarantee 3: The maximum undetectable effect (from parameters within the pre-registered values) is zero by definition.

6.4. Challenge-Period Safeguard

The six-month challenge period serves a second purpose beyond public accountability: it recruits opponents as algorithm auditors. Any party who suspects the pre-registered parameters are partisan-motivated can challenge them during the comment period. If the challenge identifies a credible partisan motivation (e.g., the submitted $\alpha_{\text{county}} = 10$ is the value that minimizes Democratic seats in a specific state), the administrator must either justify the choice on technical grounds or resubmit different parameters.

This self-enforcing mechanism does not require a neutral arbitrator to evaluate each parameter choice; it relies on adversarial scrutiny to surface problems before they affect the map.

7. Conclusion

This paper answers the adversarial reframing of U.2’s parameter-sensitivity finding: the maximum partisan shift achievable by optimizing all five BISECT parameters for partisan advantage is 0.3 Democratic seats nationally.

This is a tight bound for three structural reasons: (1) parameters affect leaf-level cut quality within a fixed bisection tree, not the tree structure itself; (2) competitive-district geography is not sensitive to 1–3 tract boundary shifts in non-competitive regions; and (3) the five parameters interact on the same geometric bottleneck, producing saturation rather than amplification.

The DIA pre-registration protocol eliminates this gaming vector entirely by requiring all parameters to be fixed and publicly recorded before Census redistricting data are released. Under pre-registration, the question “could they have set parameters to favor their party?” is converted from an unprovable allegation into a verifiable cryptographic fact: either the deployed parameters match the pre-registered hash or they do not.

The broader lesson for algorithmic redistricting design is that parameter gaming is a

minor concern relative to algorithm-structure gaming (B.0’s 18-seat gap) and geographic-definition gaming (R.3’s 1.5-seat gap). The DIA should concentrate its structural constraints on algorithm pre-specification and geographic definition mandates, treating parameter pre-registration as a lower-stakes but still necessary safeguard.

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